WEEK 4 Hill Climbing search algorithm OUTPUT:

```
Iteration: 1
Current state: [3, 1, 2, 0]
Current cost: 2
Current board:
. . . Q
. Q . .
. . Q .
Q . . .

Best neighbor: [1, 3, 2, 0]
Best neighbor cost: 1
Best neighbor board:
. . . Q
Q . . .
. . Q .
. Q . .

Iteration: 2
Current state: [1, 3, 2, 0]
Current cost: 1
Current board:
. . . Q
Q . . .
. . Q .
. Q . .

Best neighbor: [1, 3, 0, 2]
Best neighbor cost: 0
Best neighbor board:
. . Q .
Q . . .
. . . Q
. Q . .

Iteration: 3
Current state: [1, 3, 0, 2]
Current cost: 0
Current board:
. . Q .
Q . . .
. . . Q
. Q . .
```

```
Goal state reached!
Final solution: [1, 3, 0, 2]
Final cost: 0
Final board:
. . Q .
Q . . .
. . . Q
. Q . .
```

WEEK 5 Simulated Annealing OUTPUT:

```
    Iteration 30: INTERMEDIATE STATE
…   State: [4, 7, 1, 0, 4, 1, 2, 4]
    Cost: 10
      0 1 2 3 4 5 6 7
    0 . . . Q . . . .
    1 . . Q . . Q . .
    2 . . . . . . Q .
    3 . . . . . . . .
    4 Q . . . Q . . Q
    5 . . . . . . . .
    6 . . . . . . . .
    7 . Q . . . . . .

    Iteration 31: INTERMEDIATE STATE
    State: [4, 7, 1, 0, 4, 1, 2, 1]
    Cost: 12
      0 1 2 3 4 5 6 7
    0 . . . Q . . . .
    1 . . Q . . Q . Q
    2 . . . . . . Q .
    3 . . . . . . . .
    4 Q . . . Q . . .
    5 . . . . . . . .
    6 . . . . . . . .
    7 . Q . . . . . .

    Iteration 32: FINAL GOAL STATE
    State: [0, 4, 7, 5, 2, 6, 1, 3]
    Cost: 0
      0 1 2 3 4 5 6 7
    0 Q . . . . . . .
    1 . . . . . . Q .
    2 . . . . Q . . .
    3 . . . . . . . Q
    4 . Q . . . . . .
    5 . . . Q . . . .
    6 . . . . . Q . .
    7 . . Q . . . . .

    Goal state reached!
    All 8 queens are safe!
```

```
Iteration 4: INTERMEDIATE STATE
State: [0, 1, 5, 0, 0, 7, 5, 5]
Cost: 8
  0 1 2 3 4 5 6 7
0 Q . . Q Q . . .
1 . Q . . . . . .
2 . . . . . . . .
3 . . . . . . . .
4 . . . . . . . .
5 . . Q . . . Q Q
6 . . . . . . . .
7 . . . . . Q . .

Iteration 5: INTERMEDIATE STATE
State: [0, 1, 5, 6, 0, 7, 5, 5]
Cost: 7
  0 1 2 3 4 5 6 7
0 Q . . . Q . . .
1 . Q . . . . . .
2 . . . . . . . .
3 . . . . . . . .
4 . . . . . . . .
5 . . Q . . . Q Q
6 . . . Q . . . .
7 . . . . . Q . .

Iteration 6: INTERMEDIATE STATE
State: [0, 1, 5, 6, 0, 7, 4, 5]
Cost: 6
  0 1 2 3 4 5 6 7
0 Q . . . Q . . .
1 . Q . . . . . .
2 . . . . . . . .
3 . . . . . . . .
4 . . . . . . Q .
5 . . Q . . . . Q
6 . . . Q . . . .
7 . . . . . Q . .
```

```
Iteration 1: INITIAL STATE
State: [5, 4, 5, 0, 0, 7, 4, 5]
Cost: 9
  0 1 2 3 4 5 6 7
0 . . . Q Q . . .
1 . . . . . . . .
2 . . . . . . . .
3 . . . . . . . .
4 . Q . . . . Q .
5 Q . Q . . . . Q
6 . . . . . . . .
7 . . . . . Q . .

Iteration 2: INTERMEDIATE STATE
State: [5, 1, 5, 0, 0, 7, 4, 5]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . . Q Q . . .
1 . Q . . . . . .
2 . . . . . . . .
3 . . . . . . . .
4 . . . . . . Q .
5 Q . Q . . . . Q
6 . . . . . . . .
7 . . . . . Q . .

Iteration 3: INTERMEDIATE STATE
State: [0, 1, 5, 0, 0, 7, 4, 5]
Cost: 7
  0 1 2 3 4 5 6 7
0 Q . . Q Q . . .
1 . Q . . . . . .
2 . . . . . . . .
3 . . . . . . . .
4 . . . . . . Q .
5 . . Q . . . . Q
6 . . . . . . . .
7 . . . . . Q . .
```